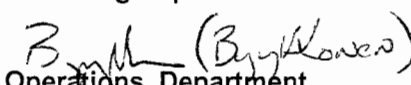


OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	Salem Generating Station		
SYSTEM:	Radioactive Waste Liquid		
TASK:	Perform an Authorized Release of Radioactive Liquid Waste to the Circulating Water System (and respond to High Radiation with failure of 2WL51 to automatically shut.)		
TASK NUMBER:	N0685140104		
JPM NUMBER:	11-01 NRC Sim h		
ALTERNATE PATH:	☒	K/A NUMBER:	068 A4.03
APPLICABILITY:		IMPORTANCE FACTOR:	
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator / Perform		
REFERENCES:	S2.OP-AB.RAD-0001, Rev. 29 (Rev checked 08-06-12) Abnormal Radiation S2.OP-SO.WL-0001, Rev. 25, (Rev checked 08-06-12) Release of Radioactive Liquid Waster from 21 CVCS Monitor Tank		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	8 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	08-06-12
Validated By:	Mulford SME or Instructor	Date:	08-23-12
Approved By:	 Training Department	Date:	9/12/12
Approved By:	 Operations Department	Date:	9/5/12
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Radioactive Waste Liquid

TASK: Perform an Authorized Release of Radioactive Liquid Waste to the Circulating Water System (and respond to High Radiation w/ failure of WL51 to automatically shut.)

TASK NUMBER: N0685140104

SIMULATOR SETUP IC-258
Have S2.OP-SO.WL-0001 open on U desk.
MALF: RM0235C Process Rad Mon 2R18 fails anywhere (1E5) tied to ET-3 with 30 second delay
MALF: VL0305 2WL51 Fails to position (0-100%) tied to ET-3 with 5 second ramp.
ET-3 kao07pnu 2WL51 open PB
ET-5 kao07tcu 2WL51 close PB (DMF VL0305)

INITIAL CONDITIONS:

Unit 2 is operating at 100% power.

An authorized radioactive liquid release from 21 CVCS Monitor Tank through 21 CCHX to Unit 1 Circ Water is being aligned in the field. The RWO performing the release is currently beginning Step 5.5, Release of 21 CVCS MT via SW System to CW System.

INITIATING CUE:

You are the Unit 2 Reactor Operator. Respond to all indications and alarms.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Operator opens 2WL51 to commence the Liquid Release.
2. Operator shuts 2WL51 after receipt of Hi Radiation Alarm on 2R18.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Waste Liquid

TASK: Perform an Authorized Release of Radioactive Liquid Waste to the Circulating Water System

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Note to Evaluator: The RWO has the official copy of the procedure in the field.		
			Evaluator: Initiate JPM by giving the following CUE: Cue: This is the RWO. Open 2WL51, Liquid Release Stop Valve IAW step 5.5.2 of S2.OP-SO.WL-0001.		
	5.5.2	Direct Unit 2 NCO to OPEN 2WL51, LIQUID RELEASE STOP VALVE.			
*			Depresses open pushbutton for 2WL51 LIQUID RELEASE STOP VALVE and verifies red open light illuminates. Simulator Operator: Ensure ET-3 is TRUE when the operator depresses 2WL51 open PB. This inserts the 2R18 High Radiation condition after a 30 second delay.		
(* If it applies here)			Announces OHA A-6, RMS HI RAD OR TRBL, as unexpected, and refers to S2.OP-AR.ZZ-0001, Overhead Annunciators - Window A. OR Checks CRT and reports CRT point 725, RMS Hi Rad or Trbl-Liq Waste Disposal CH 2R18 in alarm, and after observing 2WL51 open, shuts 2WL51 using Step 5.5.9 of S2.OP-SO.WL-0001 or S2.OP-AR.ZZ-0001 for guidance.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Waste Liquid

TASK: Perform an Authorized Release of Radioactive Liquid Waste to the Circulating Water System

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Evaluator Note: The 2WL51 status control should be maintained by the release procedure. Once the 2WL51 is shut, the Control Room operator must contact the RWO to inform them of the 2WL51 status.		
			<u>IF</u> the operator calls the RWO to report 2R18 status (on Panel 104) OR states that they would go to rack area of Control Room to verify 2R18 reading, <u>THEN</u> cue as follows as required: Cue: The RWO reports 2R18 reading 1×10^5 cps. Cue: The 2R18 in the rack area reads 1×10^5 cps.		
			Evaluator Note: <u>IF</u> the operator shuts the 2WL51 based on the Hi Radiation signal, without reporting which procedure directs the action, <u>THEN</u> allow the operator to continue until they justify their action based on procedure direction. Once the 2WL51 is shut, its status communicated to the RWO, and procedure direction for shutting the 2WL51 identified, terminate the JPM.		
			Evaluator Note: Steps for AB.RAD continue on next page.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Waste Liquid

TASK: Perform an Authorized Release of Radioactive Liquid Waste to the Circulating Water System

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Checks CRT and reports CRT point 725, RMS Hi Rad or Trbl-Liq Waste Disposal CH 2R18 in alarm.		
	ARP 3.18	<u>IF</u> 2R18 indicates Hi Radiation or trouble, <u>THEN GO TO</u> S2.OP-AB.RAD-0001, Abnormal Radiation.	Enters S2.OP-AB.RAD-0001, Abnormal Radiation.		
	AB.RAD 3.1	DETERMINE affected Radiation Monitor.	Verifies 2R18 is the affected rad monitor.		
	AB.RAD 3.2	Is the alarm, warning, or rising indication valid? (EVALUATE the following): • Chemistry Sampling • Rising activity on related monitors • Maintenance activities in vicinity of radiation detector • Transfer of radioactive resins or other materials in affected area • No indication of failure or fault on affected Radiation Monitor • Radioactive release from Salem or Hope Creek	Determines alarm must be considered valid since a radioactive liquid release has just commenced and the 2R18 is sampling that release, and the monitor has been proven operable during performance of Attachment 2, Section 2.3, which performed a source check on 2R18.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Waste Liquid

TASK: Perform an Authorized Release of Radioactive Liquid Waste to the Circulating Water System

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	AB.RAD 3.6	Announce the following on Plant PA system: - Affected Radiation Monitor number and name - Location of Radiation Monitor area with elevated indication	Announces "Radiation Monitor 2R18 Liquid Waste Disposal is in alarm, Auxiliary Building 84 elevation" Note: Operator may also include 64' elevation.		
	AB.RAD 3.7	<u>IF</u> a release to the environment is in progress, THEN DIRECT Shift Radiation Protection Technician (SRPT) to complete and provide Page 2 of the Station Status Checklist (SSCL) for release rate determination.	If the operator has NOT shut the 2WL51, THEN the operator directs the Shift Radiation Protection Technician (SRPT) to complete and provide Page 2 of the Station Status Checklist (SSCL) for release rate determination. Cue if required: IF required, state the CRS will contact the SRPT to complete page 2 of SSCL. Note: This step may not be performed based on the operators knowledge of the ECG and that the SSCL looks at Gaseous Releases.		
	AB.RAD 3.8	NOTIFY SM/CRS to REFER to Event Classification Guide, ODCM, and Technical Specifications.	Directs SM/CRS to refer to Event Classification Guide, ODCM, and Technical Specifications		
	AB.RAD 3.9	INITIATE the applicable attachment for affected Radiation Monitor:	Initiates Attachment 1 of S2.OP-AB.RAD-0001, Abnormal Radiation.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Waste Liquid

TASK: Perform an Authorized Release of Radioactive Liquid Waste to the Circulating Water System

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
(* If it applies here)	AB.RAD Att. 1 5.0	<u>IF</u> 2R18, Liquid Waste Disposal, is the affected monitor, <u>THEN</u>: 5.1 ENSURE 2WL51, TO CIRC WTR DISCHARGE, is CLOSED to stop all Liquid Waste releases. 5.2 TERMINATE Liquid Waste Release procedure in effect.	Shuts (or checks shut) 2WL51, TO CIRC WTR DISCHARGE Contacts RWO and directs them to terminate release procedure in effect.		
			Once the 2WL51 is shut, its status communicated to the RWO, and procedure direction for shutting the 2WL51 identified, terminate the JPM.		

Terminating Cue: None

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- M 1. Task description and number, JPM description and number are identified.
- M 2. Knowledge and Abilities (K/A) references are included.
- M 3. Performance location specified. (in-plant, control room, or simulator)
- M 4. Initial setup conditions are identified.
- M 5. Initiating and terminating Cues are properly identified.
- M 6. Task standards identified and verified by SME review.
- M 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- M 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 25 Date 8/23/12
- M 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- N/A 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor:  N. M. Ford Date: 8/23/12

SME/Instructor: _____ Date: _____

SME/Instructor: _____ Date: _____

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

INITIAL CONDITIONS:

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